

7. THIRD WEEK OF EMBRYONIC DEVELOPMENT

During the 3rd week, several phenomena occur:

- Gastrulation: formation of the three final germ layers.
- The beginning of two major phenomena: neurulation and metamerization.
- The development of the annexes; allantois and placenta with the formation of extra-embryonic vascular primordia and the appearance of germinal stem cells.

1.1. GASTRULATION

Gastrulation can be defined as the process by which cellular material from the ectophyll, previously superficial, migrates deeper, differentiating into a third embryonic germ layer: the mesoderm and the bilaminar embryonic disc become trilaminar. Gastrulation follows these stages:

1.1.1. Formation of the primitive streak

On the 16th day, on the upper face of the embryonic disc, in its caudal half, the cells of the ectophyll converge to form a longitudinal and median groove: this is the primitive streak.

1.1.2. Formation of the Hensen node

On the 17th day, the end of the LIVE, located roughly in the middle of the embryonic plate, ends in a thickening: Hensen's node.

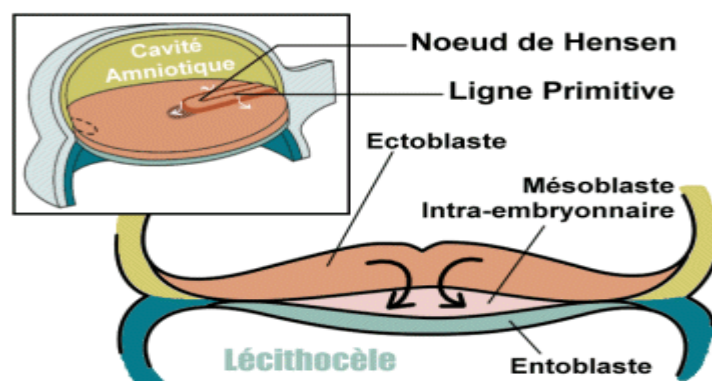


Figure 1. Invagination of ectophyllic cells and formation of the primitive line

These two structures are the starting point for a significant flow of cell migration to form a third middle layer: the mesoblast. Ectophyll cells insinuate themselves between the pre-existing ectophyll and the entophyll to constitute the mesoderm. Once the third layer is established, ectophyll and entophyte become ectoderm and endoderm, respectively.

The primitive streak determines the craniocaudal axial orientation of the embryo and its bilateral symmetry.

The mesoblast has two components:

- the lateral mesoblast which originates from the primitive streak
- the axial or chordal mesoblast originating from Hensen's node.

At both ends of the long axis of the embryonic disc, the ectoderm and endoderm remain joined, without the interposition of mesoderm; these two circular zones constitute the oopharyngeal membrane anteriorly and the cloacal membrane posteriorly.

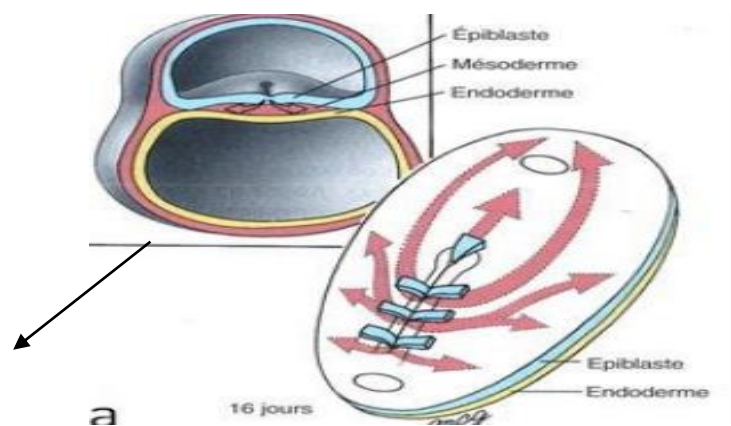


Figure 2. Gastrulation and morphogenetic movements

2. NOTOCHORDAL PROCESS

The installation of the rope involves several steps:

2.1. Formation of the cephalic process and the notochordal canal (18th and 19th day)

- From Hensen's node, the cells migrate towards the rostral end until they are close to the prochordal plate.
- They form, between the ectoderm and the endoderm, an axial diverticulum in the shape of a glove finger, the cephalic extension which extends into a canal: the notochordal canal.

2.2. Formation of the chordal groove and chordal plate (20th and 21st day)

- The floor of the notochordal canal fuses with the endoderm, and the whole structure is fissured in several places and disappears. The rest of the canal then appears as an inverted gutter; this is the notochordal gutter stage, which, while progressing

towards the caudal part, flattens to form a longitudinal plate, the notochordal plate, replacing the endoderm on the axial part of the roof of the second luteal segment.

- The disappearance of the notochordal floor opens a communication between the amniotic cavity and the secondary yolk sac at the level of Hensen's node: the Lieberkühn canal or neurenteric canal. This communication is transient and closes quickly.

2.3. Formation of the notochord (22nd day)

The definitive endoderm closes, individualizing within the embryonic mesoderm a solid chord, the notochord.

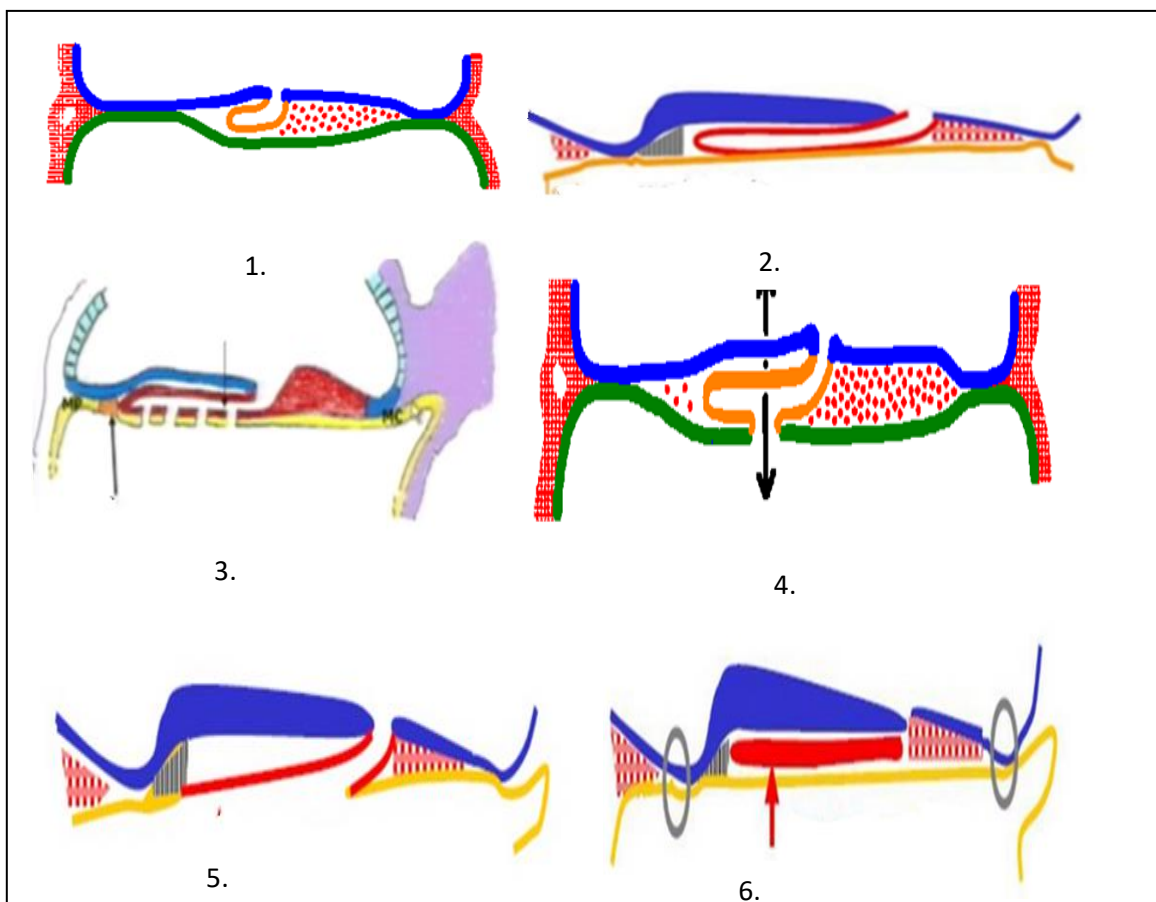
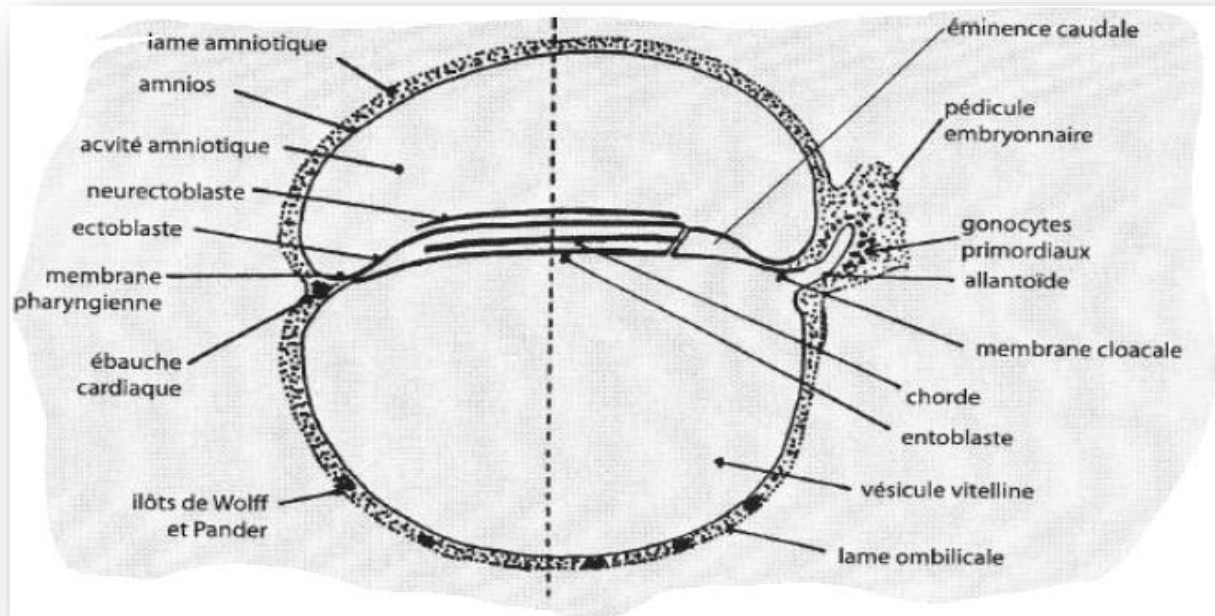


Plate 1. The stages of the notochordal process



**Figure 3. Longitudinal section of an embryo
aged 22 days, full notochord stage**

3. EMERGENCE OF CERTAIN DRAFTS AND DEVELOPMENT OF THE APPENDICES

Even before their final implementation is complete, the sheets are already beginning to undergo modifications, which will continue into the 4th week.

3.1. Neurulation

- The basic formation of the nervous system begins on the 18th day of embryonic development.
- This phenomenon is observed in cross-sections and comprises the following phases:

3.1.1. Formation of the neural plate (18th day)

- The ectoderm gives rise to two distinct cell sets, the neuroectoblast (neural plate) and the epiblast (the rest of the ectoderm).
- The neuroectoderm thickens anterior to Hensen's node and takes the form of a racket swollen anteriorly. This is the neural plate which gradually extends towards the caudal part.

3.1.2. Formation of the neural groove (19th day)

- The lateral edges of the neural plate rise up, transforming the plate into a neural groove.
- The junction zones between the edges of the neural groove and the ectoblast then

constitute two longitudinal crests: The neural crests.

3.1.3. Formation of the neural tube (21st day)

- The edges of the gutter meet and fuse in the central part of the embryonic plate.
- The closure extends towards the cephalic (faster) and caudal extremities.
- At the time of the fusion of the edges of the groove to form a tube which is the neural tube, the neural crests become isolated in the underlying mesenchyme, on either side of the tube.
- The epiblast re-establishes its continuity on the surface.

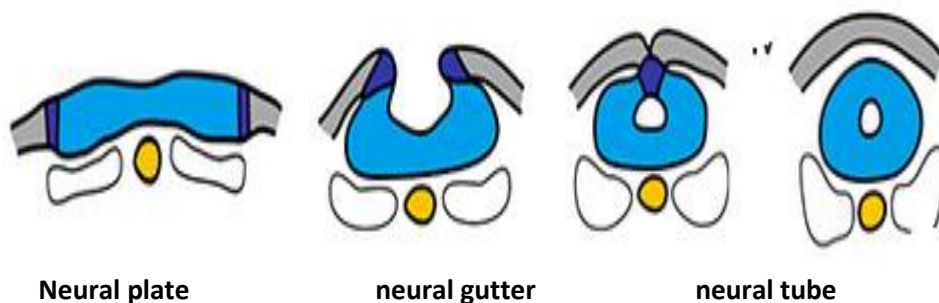


Figure 4. The stages of neurulation

3.2. Metamerization

The mesoblast (originating from the primitive streak) develops very actively from the 19th to the 21st day. Three longitudinal bands become denser within a mesenchymal tissue, on each side of the notochord: paraxial mesoblast, intermediate mesoblast, and lateral mesoblast.

3.2.1. Para-axial mesoderm

- The cells group together in large clusters which isolate themselves from each other in the longitudinal direction.
- Each cluster is called a somite.
- Segmentation begins at the rostral part of the embryonic plate (future occipital region) and progresses symmetrically towards the caudal region.

3.2.2. Intermediate mesoderm

It forms a cord which also undergoes longitudinal metameric segmentation, the nephrogenic cord, and will in particular be at the origin of the urinary system.

3.2.3. Lateral mesoderm

- It splits and gives rise to 2 layers: the intra-embryonic somatopleure and the intra-embryonic splanchnopleure.
- Between these two layers, the space is open to the intra-embryonic coelom: future peritoneo-pleuro-pericardial cavity (whose partitioning will give rise to the peritoneal cavities, pleural and pericardial).

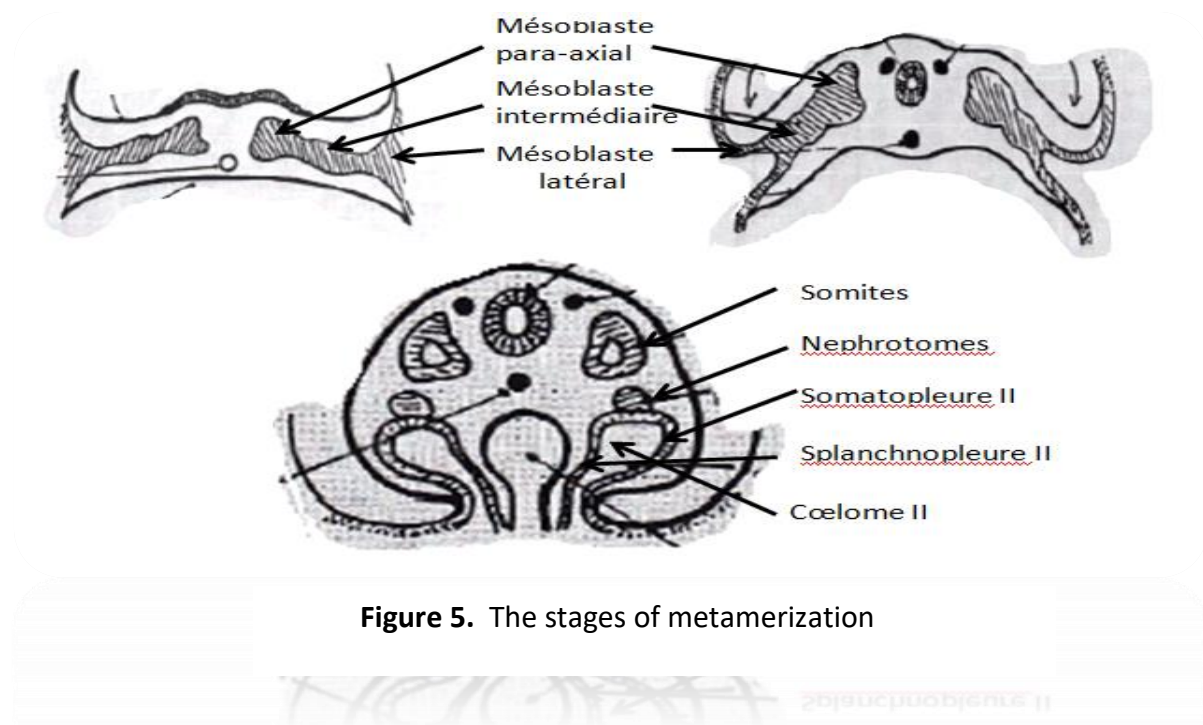


Figure 5. The stages of metamerization

3.3. Appearance of cardiovascular primordia

The first vascular primordia appear around day 17 in the extra-embryonic mesenchyme.

- Part of the mesoderm (originating from the primitive streak) migrates towards the anterior part of the embryonic disc, beyond the pharyngeal membrane, and merges with the extra-embryonic mesenchyme (cardiogenic zone). This zone is the site of the cardiac primordium, visible from day 19.
- Islands of angiogenic cells appear in the extra-embryonic mesenchyme (angiogenic zone):
 - in the wall of the yolk sac (splanchnopleure), it is the true Wolff and Pander islets which will be at the origin of the vitelline circulation and the first blood cells;
 - in the embryonic stalk, around the allantois, they will be at the origin of the allantoic (or umbilical) circulation.

- in the mesenchyme of the chorion, they will be at the origin of the vascular network of the placenta, in continuity with the allantoic circulation.

3.4. Appearance of primitive germ cells

- On the 20th day, large rounded cells appear in the extra-embryonic mesenchyme around the allantois.
- These are the stem cells of future gametes (primordial germ cells) or primordial gonocytes.
- They begin to differentiate on the 18th day from the mesoderm (primitive line) in the allantoic region.
- The primordial germ cells migrate during the 5th week (27-35 days) to colonize the developing gonads. Within the gonads, they will give rise to spermatogonia and oogonia.

3.5. Allantois Formation

- Around the 18th day, the roof of the second yolk sac (endoderm) emits a diverticulum which sinks into the embryonic stalk: the allantoic diverticulum or allantois.

3.6. Formation of the placenta

- During the 3rd week, at the same time as gastrulation takes place, the trophoblast will form placental villi which give rise to the placenta (this chapter will be covered in the embryonic annexes).